

Joining A World Wide Lab

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INTERNATIONAL
BRAIN
LABORATORY

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International Brain Laboratory

Experimental & theoretical neuroscientists collaborating to understand
brainwide circuits for complex behavior



brainhack-vandy.github.io



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Allen Institute OpenScope Leadership



Christof Koch
Meritorious Investigator
Brain & Consciousness
The Allen Institute
Co-PI on OpenScope
Award



Jerome Lecoq
Associate Investigator
Neural Dynamics Program
The Allen Institute
Co-PI on OpenScope award





NEWS

World's first crowd-sourced neuroscience study aims to understand how our brains predict the future

Mar 14, 2025

Using the Allen Institute's OpenScope platform, experiments will probe brain functions related to "predictive processing"



brainhack-vandy.github.io

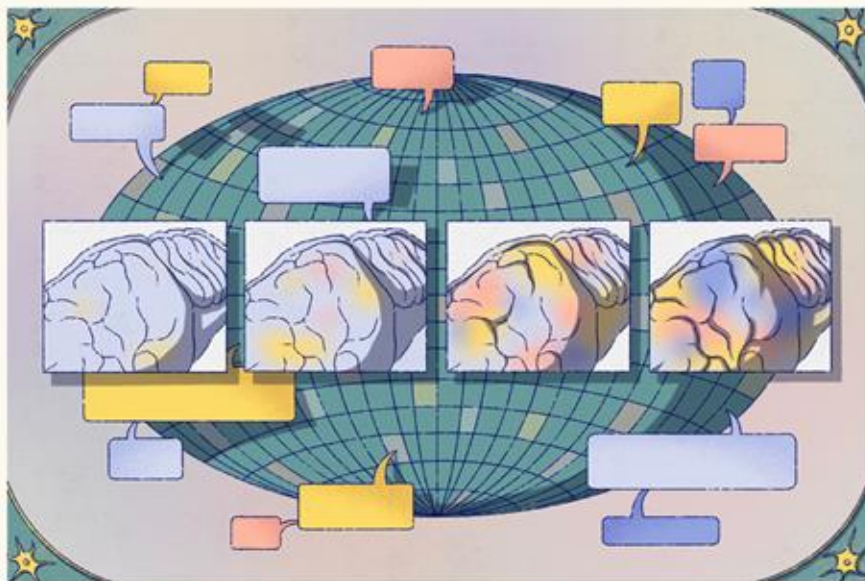
PERSPECTIVES / OPEN NEUROSCIENCE

A community-designed experiment tests open questions in predictive processing

More than 50 scientists came together to identify the key missing data needed to rigorously test theoretical models.

BY JÉRÔME LECOQ

12 NOVEMBER 2025 | 6 MIN READ



Team effort: The distributed community has helped collectively analyze pilot data collected at the Allen Institute.
ILLUSTRATION BY YIHUI CHANG

[Submitted on 13 Apr 2025]

Neural mechanisms of predictive processing: a collaborative community experiment through the OpenScope program

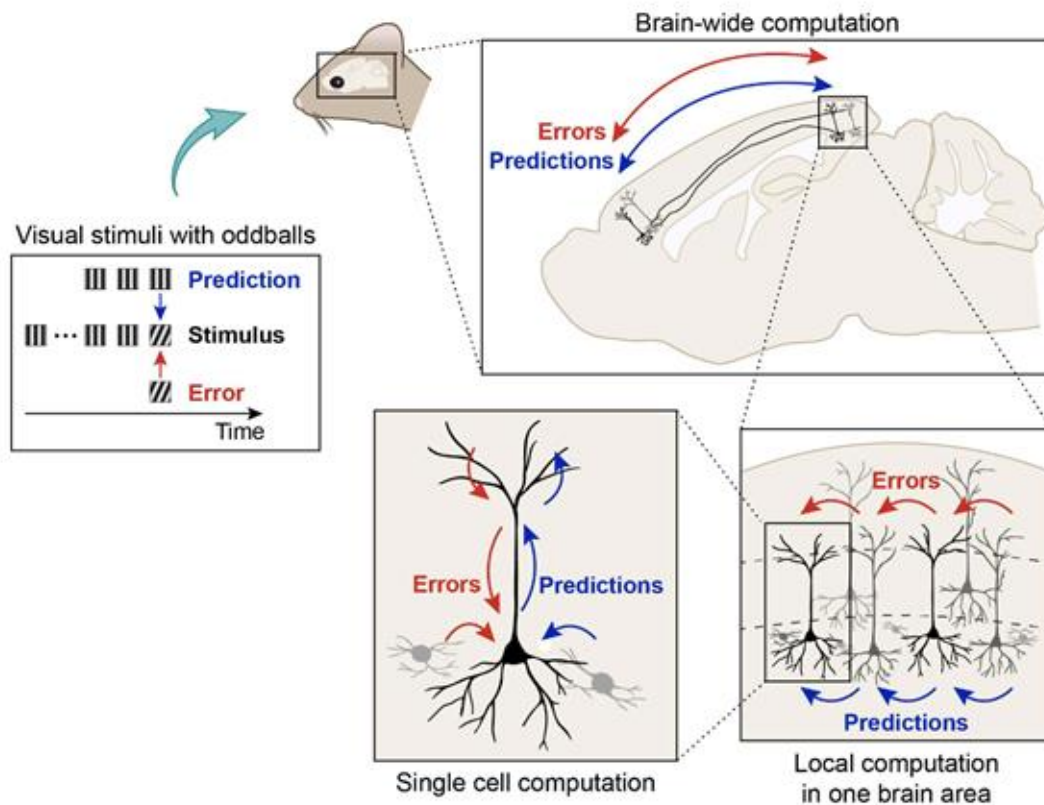
Ido Aizenbud, Nicholas Audette, Ryszard Auksztulewicz, Krzysztof Basiński, André M. Bastos, Michael Berry, Andres Canales-Johnson, Hannah Choi, Claudia Clopath, Uri Cohen, Rui Ponte Costa, Roberto De Filippo, Roman Doronin, Steven P. Errington, Jeffrey P. Gavornik, Colleen J. Gillon, Arno Granier, Jordan P. Hamm, Loreen Hertäg, Henry Kennedy, Sandeep Kumar, Alexander Ladd, Hugo Ladret, Jérôme A. Lecoq, Alexander Maier, Patrick McCarthy, Jie Mei, Jorge Mejias, Fabian Mikulasch, Noga Mudrik, Farzaneh Najafi, Kevin Nejad, Hamed Nejat, Karim Oweiss, Mihai A. Petrovici, Viola Priesemann, Lucas Rudelt, Sarah Ruediger, Simone Russo, Alessandro Salatiello, Walter Senn, Eli Sennesh, Sepehr Sima, Cem Uran, Anna Vasilevskaya, Julien Vezoli, Martin Vinck, Jacob A. Westerberg, Katharina Wilmes, Yihan Sophy Xiong

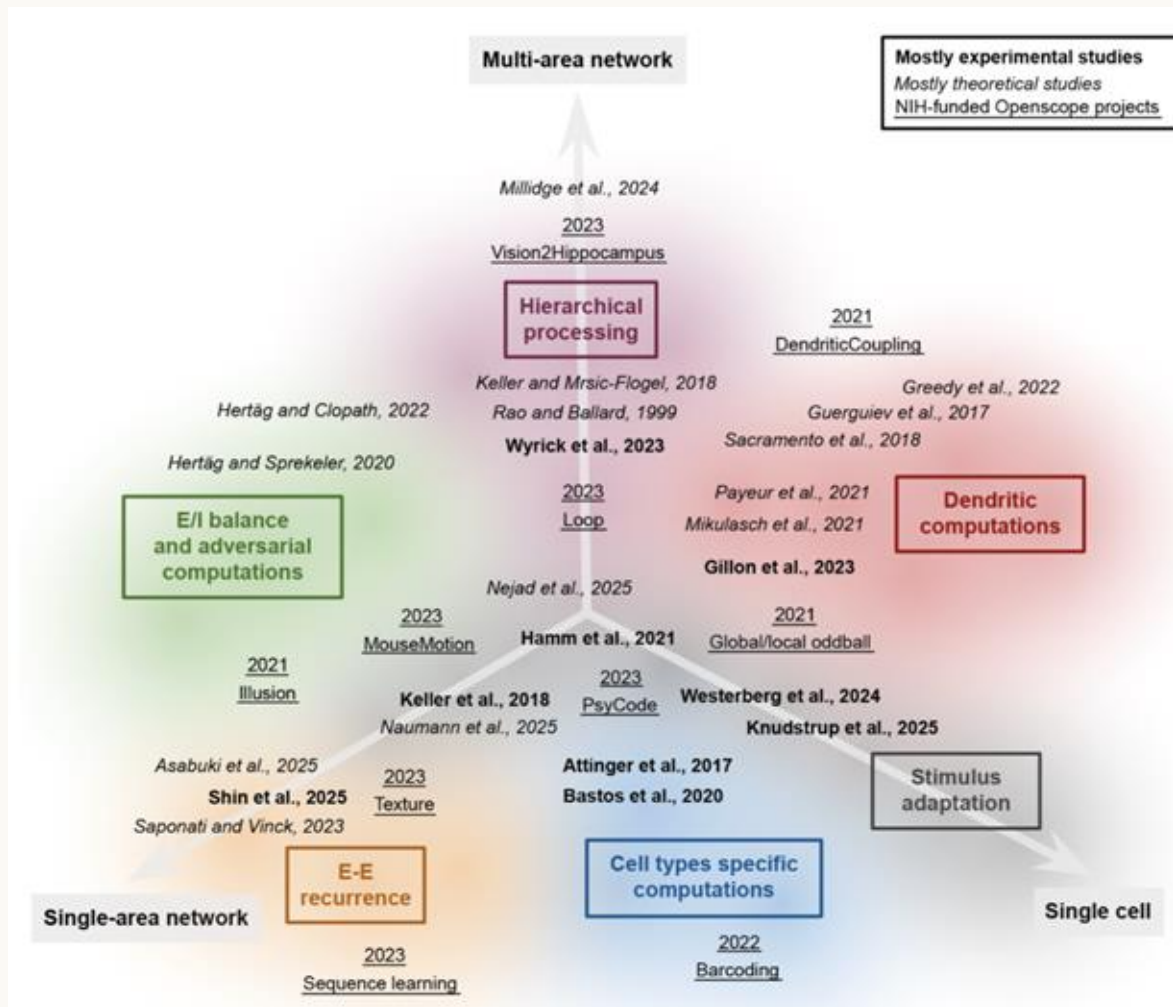
This review synthesizes advances in predictive processing within the sensory cortex. Predictive processing theorizes that the brain continuously predicts sensory inputs, refining neuronal responses by highlighting prediction errors. We identify key computational primitives, such as stimulus adaptation, dendritic computation, excitatory/inhibitory balance and hierarchical processing, as central to this framework. Our review highlights convergences, such as top-down inputs and inhibitory interneurons shaping mismatch signals, and divergences, including species-specific hierarchies and modality-dependent layer roles. To address these conflicts, we propose experiments in mice and primates using in-vivo two-photon imaging and electrophysiological recordings to test whether temporal, motor, and omission mismatch stimuli engage shared or distinct mechanisms. The resulting dataset, collected and shared via the OpenScope program, will enable model validation and community analysis, fostering iterative refinement and refutability to decode the neural circuits of predictive processing.

Subjects: **Neurons and Cognition (q-bio.NC)**

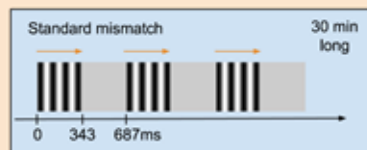
Cite as: [arXiv:2504.09614](https://arxiv.org/abs/2504.09614) [q-bio.NC]



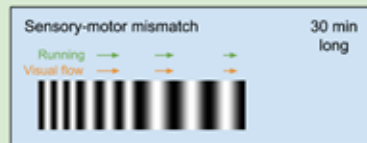




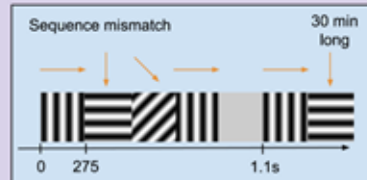
Session 1 - Standard mismatch



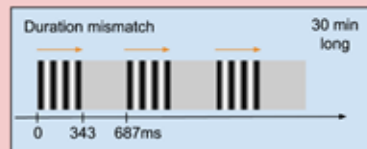
Session 2 - Sensory-motor mismatch



Session 3 - Sequence mismatch



Session 4 - Duration mismatch



RECORDING SESSIONS

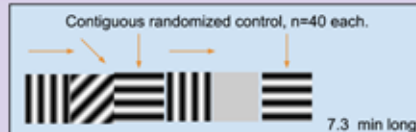
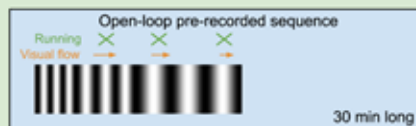
Oddballs occur randomly
Every 11s
n=40 each

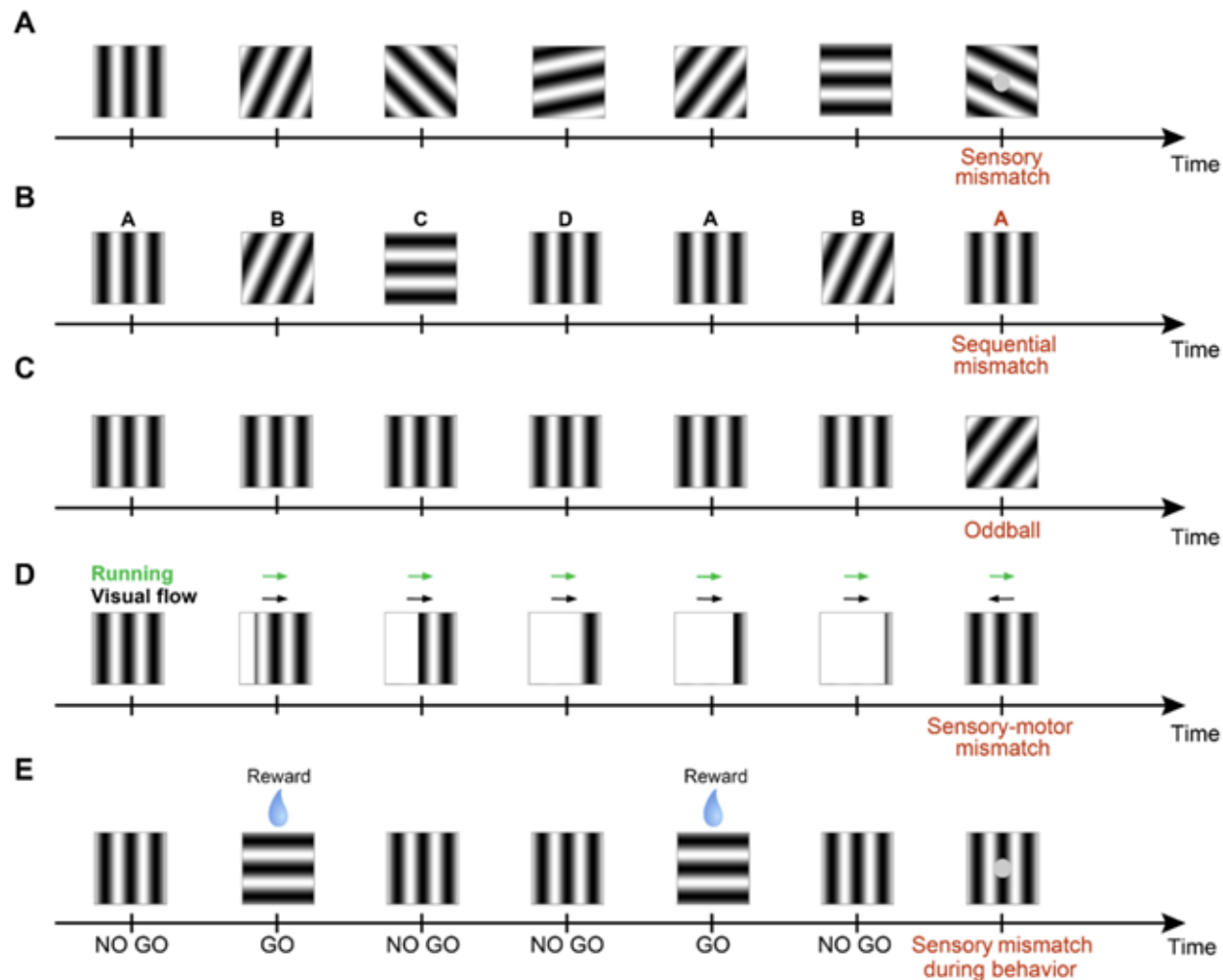


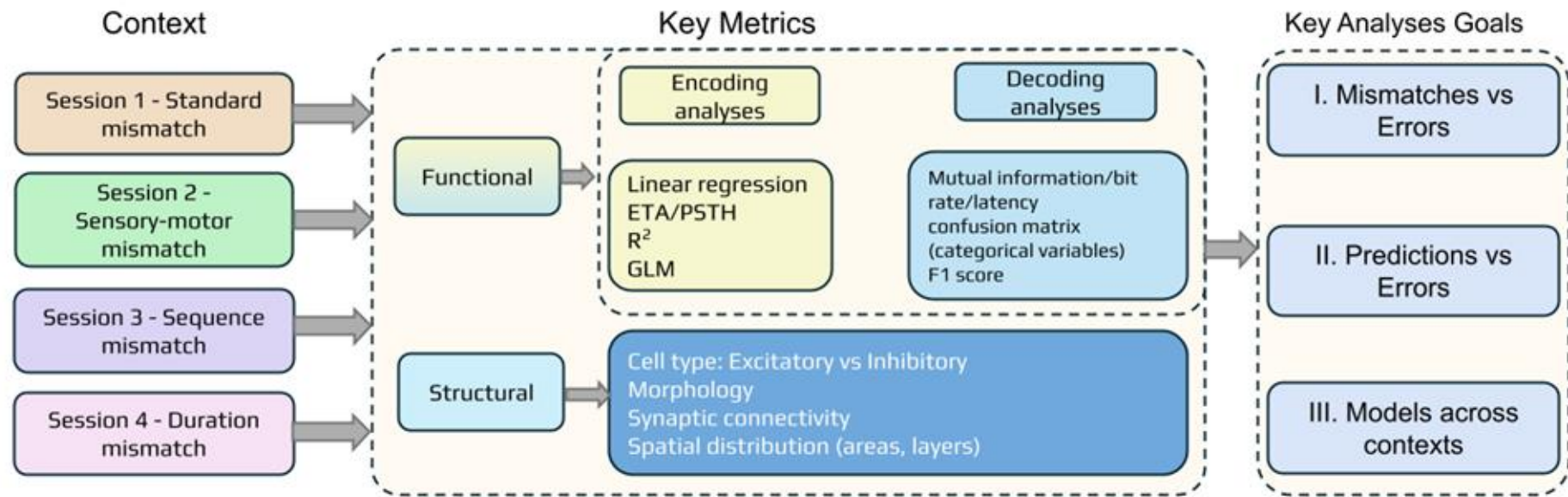
Duration oddballs

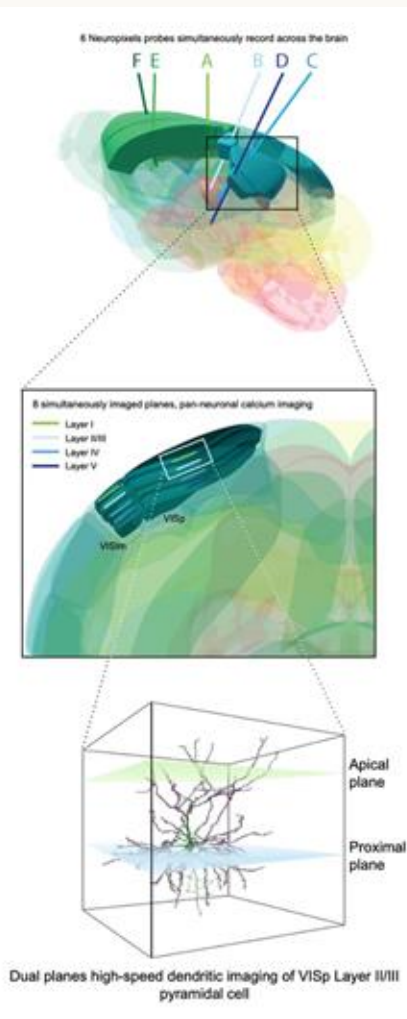


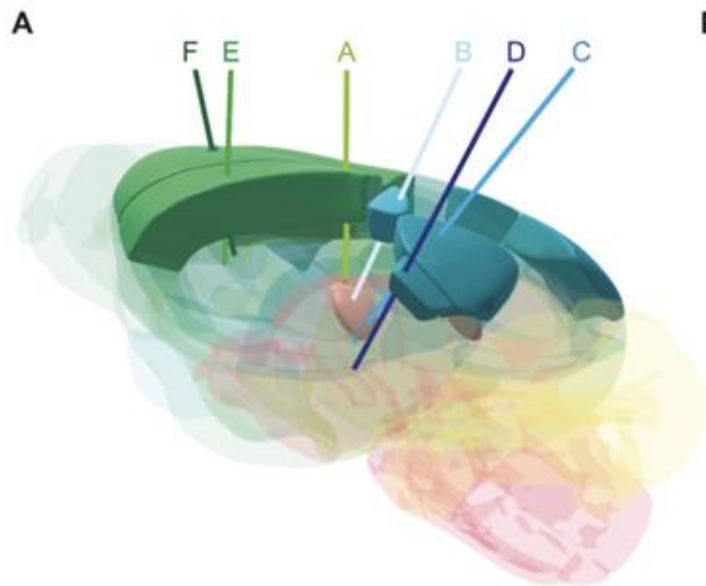
Shared control blocks

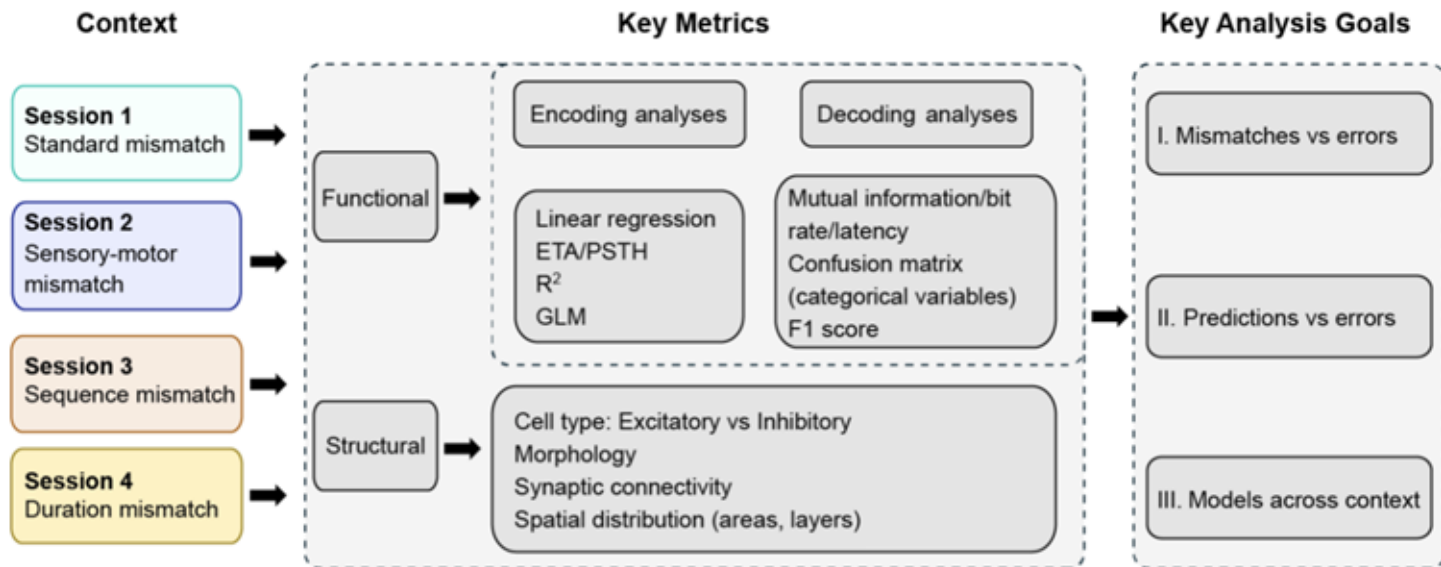












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OpenScope Community Predictive Processing

Welcome to the **OpenScope Community Predictive Processing** repository! This website serves as a hub for community-driven contributions, documentation, and project tracking for our collaborative neuroscience research initiative.

Quick Start Guide

If you are a...	Start here
New Contributor	How to Contribute and Collaboration Policy
Experimental Scientist	Experimental Plan and Hardware Overview
Data Scientist	Analysis Plan and Experiment Summary
Curious Visitor	FAQ and Project Overview below





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Breakthrough AI
to solve the world's
biggest problems



ID	Experiment Hypothesis	Surp... ↓	Belief Before	Belief After	Direction
3	The functional connectivity between the primary visual cortex (VISp) and the lateromedial visual area (VISl) is significantly stronger during periods of large pupil diameter compared to periods of small pupil diameter.	0.560	Maybe True	Maybe False	Negative
2	The amplitude of population calcium transients (dF/F) time-locked to stimulus onsets is significantly amplified during periods of locomotion compared to stationary periods in the higher-order visual area VISl.	0.487	Likely True	Maybe False	Negative
1	Running speed exhibits a stronger positive correlation with the average population calcium activity (dF/F) in the primary visual cortex (VISp) compared to the higher-order lateromedial visual area (VISl).	0.365	Maybe False	Likely False	Negative
4	The proportion of neurons significantly modulated by running speed differs across the four imaging planes of the primary visual cortex (VISp_0 through VISp_3), reflecting depth-dependent arousal inputs.	0.284	Maybe True	Likely True	Positive

Hypothesis

Belief before experiment: **Likely True (0.750)**

The amplitude of population calcium transients (dF/F) time-locked to stimulus onsets is significantly amplified during periods of locomotion compared to stationary periods in the higher-order visual area VISI.

Analysis

Belief after experiment: **Maybe False (0.438)**

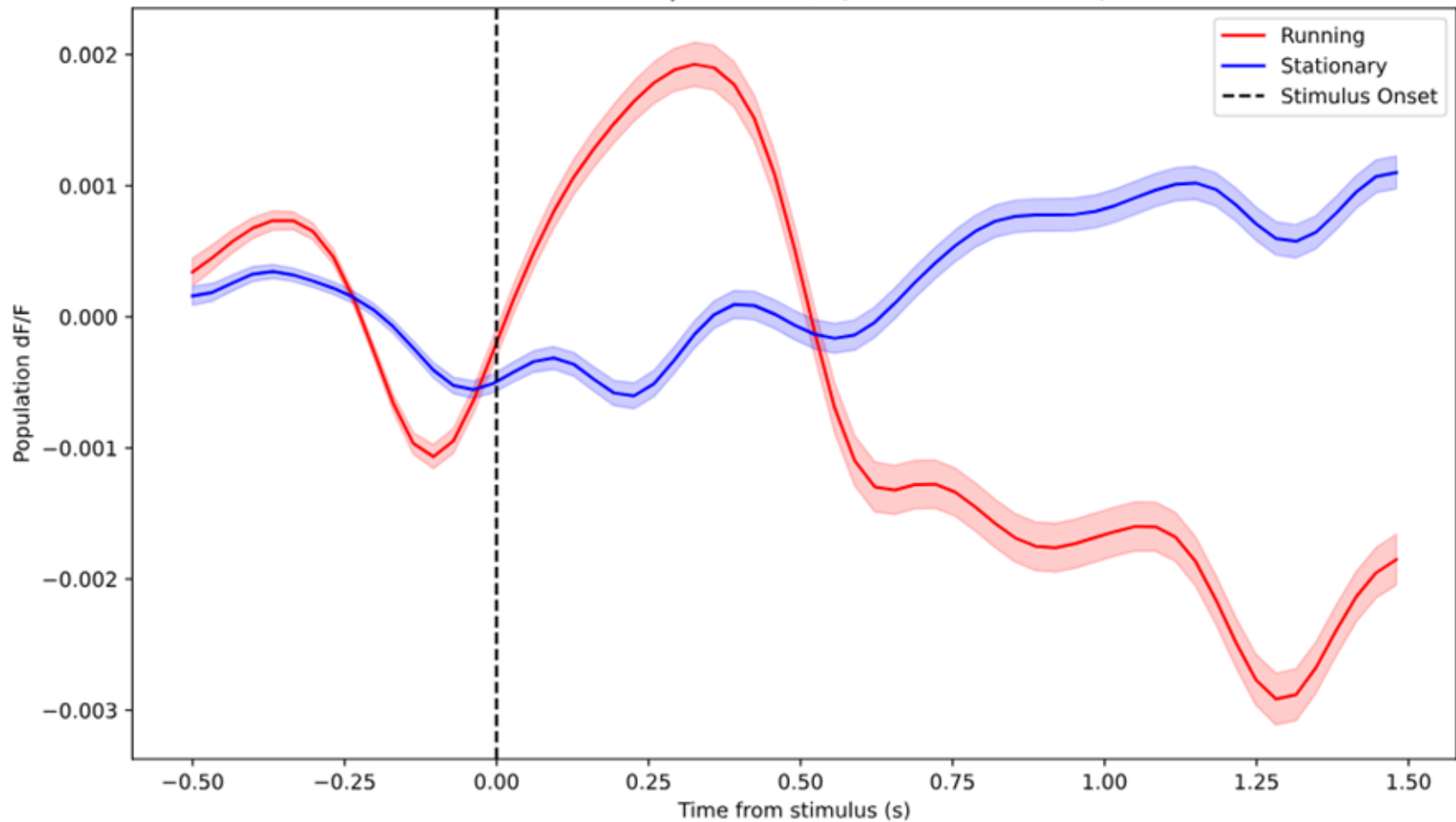
The experiment successfully loaded the multiplane ophys NWB dataset and processed the required modules. Stimulus timestamps were successfully categorized based on the subject's behavioral state, yielding 11,797 'running' trials (>2 cm/s) and 64,839 'stationary' trials (<0.5 cm/s). The code then extracted epochs of VISI population dF/F aligned to stimulus onsets and performed baseline subtraction. Statistical comparison of the peak amplitudes within the 0.0 to +1.5s window revealed a mean peak amplitude of 0.0182 ± 0.0180 for running and 0.0301 ± 0.0260 for stationary states. The independent t-test showed a highly significant difference ($t = -47.71$, $p = 0.0$), indicating that the absolute maximum amplitude across the entire 1.5s window was higher on average during stationary periods. However, the generated Peri-Stimulus Time Histogram (PSTH) highlights distinct temporal dynamics: locomotion is associated with a sharp, immediate early-phase transient response (amplified early gain) followed by suppression, whereas the stationary state evokes a delayed and gradual late-phase increase in calcium activity. Overall, the execution successfully delivered all requested outputs and statistical comparisons.

Code

```
1 import subprocess
2 import sys
3 import os
4 import warnings
5
6 def install(package):
7     subprocess.check_call([sys.executable, "-m", "pip", "install", "--quiet", package])
8
9 for pkg in ["pynwb", "scipy", "matplotlib"]:
10     try:
```

View all 195 lines 

PSTH of VISI Population dF/F (Baseline Subtracted)



I was accepted to participate in...

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July 07-13

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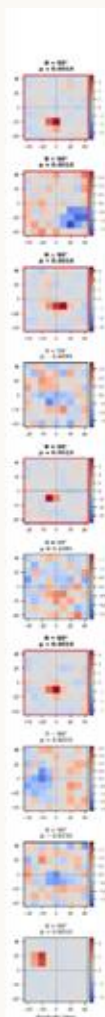


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HUMAN

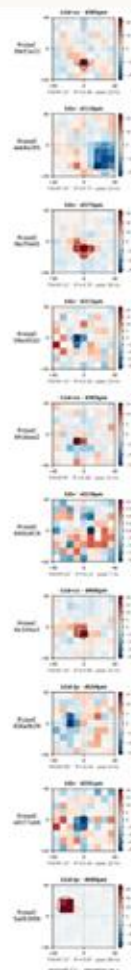


RaabeM



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